

MolTrustBench Supplementary Information

Briefings in Bioinformatics submission-hardening package, v31 audit-QC build.

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This PDF is a compact reviewer-facing guide to the machine-readable supplement package. Full CSV/JSON artifacts remain in the companion ZIP and tables directory; large audit tables are summarized here rather than expanded into hundreds of PDF pages.

Companion files: [paper/supplement/MolTrustBench_BIB_supplement_package_20260529_210256.zip](#); [paper/supplement/supplement_manifest.csv](#); [paper/supplement/tables/](#); [paper/supplement/trust_cards/](#).

QC Summary

Check	Status	Detail
Tables S1-S29 available	pass	29 manifest rows with row counts and SHA256 hashes
S24 covers core metrics	pass	16 rows from results/tables/slice_metrics.csv represented
Valid classification rows have class counts	pass	0 valid rows missing class counts
One-class AUROC rows invalid	pass	0 one-class rows marked metric-valid
Low-minority rows flagged	pass	836 boundary; 220 caution
S1 scaffold accounting	pass	cumulative first-seen scaffold counts are monotonic and per-release counts are labeled
S29 source map	pass	uses Fig. S7 for label-shuffle and current Fig. 3 exposure-delta source naming

Supplement Table Inventory

Rows, checksums, and source paths are taken from `supplement_manifest.csv`. SHA256 values are truncated visually here; full values are in the manifest.

Table	Title	Claim	Role	Rows	File	SHA256
Table_S1	ChEMBL release-index summary	C1/C2	R4/R9	5	tables/Table_S1_chembl_release_counts.csv	dfa17de16c...
Table_S2	Benchmark exposure summary across audited tasks	C1/C2	R1/R2	4	tables/Table_S2_benchmark_coverage_exposure_summary.csv	13653329b2...
Table_S3	Core BBBP/ClinTox exposure-adjusted baseline metrics	C3	R6/R8	16	tables/Table_S3_slice_metrics.csv	d11c27003f...
Table_S4	BACE/Tox21 C3 expansion metrics	C3	R6/R8	30	tables/Table_S4_bace_tox21_c3_summary.csv	7983f7fab3...
Table_S5	BACE/Tox21 sequence-family metrics	C3/C5	R6/R8	60	tables/Table_S5_bace_tox21_sequence_summary.csv	fc1d5a40fe...
Table_S6	Exposure-adjusted CI summary	C3	R2/R6	80	tables/Table_S6_exposure_delta_ci.csv	3e3ae609fa...
Table_S7	Sequence-family CI summary	C3	R2/R6	24	tables/Table_S7_sequence_delta_ci.csv	2c2cd95105...
Table_S8	ESOL/Lipophilicity regression CI summary	C3	R8	24	tables/Table_S8_regression_delta_ci.csv	c0de2c6d50...
Table_S9	Endpoint-native ChEMBL temporal metrics	C3/C4	R4/R8/R10	108	tables/Table_S9_chembl_endpoint_temporal_summary.csv	b3611bfe57...
Table_S10	Endpoint SMILES-Transformer add-on metrics	C3	R6/R8	42	tables/Table_S10_chembl_endpoint_transformer_summary.csv	ced1a2935c...
Table_S11	Train-label-shuffle null-control comparisons	C3	R2/R6	22	tables/Table_S11_label_shuffle_null_control_comparison.csv	c668fce1d7...
Table_S12	CYP label-shuffle rescue null summary	C3	R2/R6/R8	3	tables/Table_S12_cyp_extension_label_shuffle_null_summary.csv	378c3173e7...
Table_S13	Frozen MolFormer embedding summary	C3	R5/R7	57	tables/Table_S13_fm_embedding_summary.csv	80ce23b5dc...
Table_S14	Frozen ChemBERTa-100M embedding summary	C3	R5/R7	30	tables/Table_S14_chemberta_100m_fm_summary.csv	484bac6e64...
Table_S15	Assay-provenance task summary with DILI/hepatotoxicity	C4	R10	6	tables/Table_S15_assay_provenance_task_summary_with_dili.csv	49e3d40289...
Table_S16	CYP1A2/CYP2C19 assay-provenance summary	C4	R10	2	tables/Table_S16_cyp_endpoint_extension_assay_provenance_summary.csv	d8fea08cf...
Table_S17	DILI/hepatotoxicity split and provenance boundary	C4	R8/R10	6	tables/Table_S17_dili_hepatotoxicity_split_summary.csv	16de112cbf...
Table_S18	DILI/hepatotoxicity sequence summary	C3/C4	R8/R10	12	tables/Table_S18_dili_hepatotoxicity_sequence_summary.csv	36ff8bb845...
Table_S19	Model pretraining registry	C5	R5	12	tables/Table_S19_model_pretraining_registry.csv	b025998616...
Table_S20	Current reviewer controls and reporting checklist	C5	R1-R10	10	tables/Table_S20_required_controls.csv	f2ec4b8b91...
Table_S21	Trust-card example source data	C5	R6	4	tables/Table_S21_trust_card_examples.csv	f2e8b54b6d...
Table_S22	Workflow artifact map	C5	R6	7	tables/Table_S22_workflow_artifact_map.csv	627e5da3f9...
Table_S23	Expected limitation and critical-failure accounting	C3/C5	R8	6	tables/Table_S23_expected_limitations_summary.csv	adacd18c1b...
Table_S24	Slice-validity master table	C3	R8	12626	tables/Table_S24_slice_validity_master.csv	f9534654fd...
Table_S25	Standardization audit summary	C1/C2	R9	13	tables/Table_S25_standardization_audit.csv	e1a020cbe7...
Table_S26	Assay-provenance flag dictionary	C4	R10	6	tables/Table_S26_assay_flag_dictionary.csv	caa4fc575b...

Table_S27	Assay-provenance example rows	C4	R10	22	tables/Table_S27_assay_provenance_examples.csv	990ce3f3c...
Table_S28	Machine-readable trust-card schema	C5	R6	12	tables/Table_S28_trust_card_schema.csv	fa93617df1...
Table_S29	Figure source-data map	C5	R6	8	tables/Table_S29_figure_source_data_map.csv	cd6b3fd091...

Key Source Tables

Table S1. ChEMBL Release-Index Summary

Release	Date	Raw chemreps	New InChIKeys	New scaffolds	Cumulative scaffolds	Scaffold failures
CHEMBL24	2018-05	1820035	1820035	525438	525438	20572
CHEMBL27	2020-05	1941410	135127	38987	564425	915
CHEMBL30	2022-02	2136187	194983	60599	625024	1552
CHEMBL33	2023-05	2372674	236538	120287	745311	626
CHEMBL36	2025-07	2854815	482969	163644	908955	1222

Table S2. Four-Task Exposure Summary

task_name	n_molecules	exact_exposure_rate	scaffold_exposure_rate	nn_exposure_rate_08
BBBP	2039	0.8376655223148602	0.8778813143697891	0.8891613536047082
ClinTox	1480	0.4736486486486486	0.7371621621621621	0.5993243243243244
BACE	1513	0.2974223397224058	0.4269662921348314	0.3939193654990086
Tox21	7258	0.9207770735739872	0.7529622485533205	0.9585285202535132

Table S21. Trust-Card Source Values

cutoff_release	n_molecules	nn_exposure_rate_08	exact_unobserved_n
CHEMBL30	2039	0.889161	331
CHEMBL30	1480	0.599324	779
CHEMBL30	1513	0.393919	1063
CHEMBL30	7258	0.958529	575

Table S24. Slice-Validity Summary

Interpretation status	Rows
expected_limitation_invalid	6740
not_auditable_in_compact_package	2454
class_balance_not_applicable_regression	1300
low_minority_class_boundary	836
class_balance_ok	571
one_class_invalid	445
small_minority_class_caution	220
missing_class_counts	60

Full S24 is machine-readable in CSV form and includes source table, task, model, split, test n, class counts, metric-validity flag, CI availability, and reason code for each row.

Table S25. Standardization Audit Summary

Dataset	Kind	Release	Raw	Std	Unique	Exposure rows
ChEMBL release index CHEMBL24	release_index_first_seen_release	CHEMBL24	1820035	1820035	1820035	
ChEMBL release index CHEMBL27	release_index_first_seen_release	CHEMBL27	1941410	1941410	1941410	
ChEMBL release index CHEMBL30	release_index_first_seen_release	CHEMBL30	2136187	2136187	2136187	
ChEMBL release index CHEMBL33	release_index_first_seen_release	CHEMBL33	2372674	2372674	2372674	
ChEMBL release index CHEMBL36	release_index_first_seen_release	CHEMBL36	2854815	2854815	2854815	
chembl_CYP1A2_pchembl5_temporal	endpoint_native_processed_benchmark		6000	6000	6000	6000

chembl_CYP2C19_pchembl5_temporal	endpoint_native_processed_benchmark		6000	6000	6000	6000
chembl_DILI_hepatotoxicity_binary_temporal	endpoint_native_processed_benchmark		513	513	513	513
chembl_hERG_pilot	endpoint_native_processed_benchmark		5000	5000	4999	
moleculenet_BACE_exposure	real_benchmark_exposure_annotation		1513	1513	1513	1513
moleculenet_BBBP_exposure	real_benchmark_exposure_annotation		2039	2039	1971	2039
moleculenet_ClinTox_exposure	real_benchmark_exposure_annotation		1480	1480	1455	1480
moleculenet_Tox21_exposure	real_benchmark_exposure_annotation		7258	7258	7256	7258

Table S29. Figure Source-Data Map

Figure	Figure file	Source data	Supplement source	Status
Fig. 1	workflow_schematic.pdf	results/tables/workflow_artifact_map.csv	tables/Source_Fig_1_workflow_artifact_map.csv	available
Fig. 2	exposure_heatmap_coverage.pdf	results/tables/benchmark_coverage_exposure_summary.csv	tables/Source_Fig_2_benchmark_coverage_exposure_summary.csv	available
Fig. 3	exposure_delta_ci.pdf	results/tables/fig3_exposure_delta_ci_source.csv	tables/Source_Fig_3_fig3_exposure_delta_ci_source.csv	available
Fig. 4A	bace_tox21_sequence_model_family.pdf	results/tables/fig5_sequence_family_delta_source.csv	tables/Source_Fig_4A_fig5_sequence_family_delta_source.csv	available
Fig. 4B	bace_tox21_sequence_model_family.pdf	results/tables/fig5_sequence_family_comparison_size_source.csv	tables/Source_Fig_4B_fig5_sequence_family_comparison_size_source.csv	available
Fig. 5	assay_conflict_map.pdf	results/tables/assay_provenance_task_summary_with_dili.csv	tables/Source_Fig_5_assay_provenance_task_summary_with_dili.csv	available
Fig. 6	trust_card_examples.pdf	results/tables/trust_card_examples.csv	tables/Source_Fig_6_trust_card_examples.csv	available
Fig. S7	label_shuffle_null_control.pdf	results/tables/figS7_label_shuffle_source.csv	tables/Source_Fig_S7_figS7_label_shuffle_source.csv	available

Data Availability / Deposit Notes

data_availability_statement.md

Data Availability Statement For BIB Submission

The data availability statement below is ready to paste after replacing the repository placeholder with a DOI or accession.

Ready-To-Paste Draft

All processed data, exposure annotations, generated benchmark splits, summary tables, figure source data, supplementary tables, trust-card JSON examples, and analysis manifests needed to reproduce the results in this article are provided in the MolTrustBench evidence archive and supplement package: `**[repository DOI/accession to be inserted]**`. The local archive corresponding to this draft is ``results/archive/moltrustbench_data_freeze_20260529_161700.tgz`` (SHA256 ``beb779864151105b3c8cd13507df8f0779bfc53e2d6a4f6a792d82f91e34b2df``; manifest ``moltrustbench_data_freeze_20260529_161700_manifest.csv``). The supplement package under ``paper/supplement/`` includes Tables S1-S29, including slice validity, standardization audit, assay-flag definitions, provenance examples, trust-card schema, and figure source-data mapping. The archive excludes raw ChEMBL database dumps, local secrets, transient caches, and third-party model checkpoints. ChEMBL source releases are publicly available from the official ChEMBL downloads page, and benchmark datasets are available from their original public sources. Frozen foundation-model checkpoint artifacts are not redistributed in the archive; scripts record the staged checkpoint identifiers and reproduce embeddings when the corresponding public checkpoint is available under its own license. Code is available at the project repository `**[GitHub URL and commit SHA to be inserted]**`.

Repository Actions Before Submission

- * Deposit the current data-freeze archive, manifest, and BIB supplement package in a DOI-issuing repository such as Zenodo, Figshare, OSF, or Harvard Dataverse.
- * Add the DOI/accession to the statement above and to ``paper/latex/moltrustbench_main.tex``.
- * Add the GitHub commit SHA or tagged release corresponding to the submitted manuscript.
- * Include dataset citations in the reference list for ChEMBL and any third-party benchmark datasets used in the article.

Placeholder Search Before Submission

- * ``[repository DOI/accession to be inserted]``
- * ``[GitHub URL and commit SHA to be inserted]``
- * ``DOI added during production``
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- * ``submission-metadata-required@example.invalid``
- * author, affiliation, funding, acknowledgment, and conflict-of-interest placeholders

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deposit_readme.md

MolTrustBench Deposit README

Purpose: make the Briefings in Bioinformatics submission auditable without redistributing raw ChEMBL dumps or third-party checkpoints.

Upload To DOI Repository

Upload these artifacts together to Zenodo, Figshare, OSF, or Harvard Dataverse:

- * ``results/archive/moltrustbench_data_freeze_20260529_161700.tgz``
- * ``results/archive/moltrustbench_data_freeze_20260529_161700_manifest.csv``
- * ``results/archive/moltrustbench_data_freeze_20260529_161700_summary.json``
- * latest ``paper/supplement/MolTrustBench_BIB_supplement_package_*.zip``
- * final Overleaf/OUP LaTeX bundle from ``paper/latex_compile_bundle/``
- * final compiled manuscript PDF, if the repository allows manuscript previews
- * tagged GitHub code snapshot or a Zenodo-linked GitHub release archive

Include In The Deposited README

- * Archive SHA256:
``beb779864151105b3c8cd13507df8f0779bfc53e2d6a4f6a792d82f91e34b2df``
- * GitHub repository URL and release tag or commit SHA.
- * DOI/accession assigned by the repository.
- * License for generated evidence tables and code snapshot.
- * Access date for ChEMBL releases, benchmark datasets, and public checkpoints.
- * Python, RDKit, pandas, scikit-learn, PyTorch, and plotting environment notes.
- * Command to regenerate supplement tables:
``python scripts/make_bib_supplement_package.py``
- * Command to rerun paper-facing sanity checks:
``python -m moltrustbench.evaluation.sanity_checks --fail-on-critical``
- * Statement that ChEMBL release dates are public-observability dates, not model-specific pretraining dates.

Do Not Upload

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- * ``.env``, ``.env.local``, SSH keys, tokens, or remote host credentials;
- * Hugging Face or other public model checkpoints unless the license explicitly permits redistribution;
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Before submission, replace placeholders in:

- * ``paper/latex/moltrustbench_main.tex``
- * ``paper/data_availability_bib.md``
- * ``paper/supplement/data_availability_statement.md``
- * ``paper/supplement/README.md``, if a DOI-specific note is added

Search these exact placeholders and replace them only after deposit/release:

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- * ``DOI added during production``
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- * ``Author One``, ``Author Two``, ``MolTrustBench Consortium``
- * ``author@example.com``
- * funding, acknowledgment, conflict-of-interest, and affiliation placeholders

Use DataCite-style citations for deposited evidence and cite ChEMBL, benchmark datasets, and public checkpoints through their original sources.